

Sleep No Mosquitoes: Compensating Distance-Aware Compression in a VR Audio Game

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ABSTRACT

Can you catch mosquitoes at night? Sleep No Mosquitoes is a VR game with audio as the main interactive element. The intention is to explore how to compensate for the problem of distance perception compression in the VR environment through interesting mechanisms. The game is divided into entertainment mode and experimental mode, inviting 21 players and 10 experimental participants to test the factors that affect distance perception in VR. According to the analysis combining quantitative and qualitative methods, we found that three methods - dynamic head movement, dynamic audio, and sound effects of body movement - can improve players' efficiency in catching mosquitoes, and effectively compensate for the distortion of distance perception in VR.

CCS CONCEPTS

• **Human-centered computing** → *Virtual reality*; **User interface design**.

KEYWORDS

Virtual Reality; Virtual audio game; Distance Compression; Game Design

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1 INTRODUCTION

In the broader context of HCI, audio has received considerable attention for enhancing a range of backgrounds through the sonification of interfaces and information [27, 30]. Particularly in VR, which provides a unique interaction by offering a more intuitive interaction than typical desktop computer environments, it has become an important scene for audio research and expanded the application of audio in immersive games [15]. In 3D audio design for VR, users primarily use head movement to help locate audio in virtual space [31], which enhances the sense of "motion-acoustic

consistency" between body movements and resulting audio cues [7]. Spatial audio localization is a common design approach in VR audio design.

In fact, audio-rich development modes are often overlooked in the development of various media projects [25]. With the increasing design requirements, VR needs to improve its perception accuracy of spatial scale and depth. This presents a challenge in modern VR experiences, known as distance distortion perception [4, 10, 29], also known as distance compression, where users perceive distances in VR as closer than in real space [24]. Some researchers have shown that the perceived self-centered distance (distance from the observer to the object) in virtual environments is only 74% of the actual distance [28]. This inaccurate perception of distance can also affect users' judgments of size, height, scale, speed, and other related information [9]. Spatial distance perception is strongly related to spatial audio localization, but it is not the same concept. Despite this challenge, we rarely find relevant design methods at the HCI and VR game level.

As a response to this research challenge, we propose a game design as a probe to explore the positive effects of using HRTF(Head-Related Transfer Function) to compensate for distance compression in VR. In the VR rhythm game "Sleep no mosquitoes," we designed three levels where players catch mosquitoes, creating a competitive and motivating game atmosphere to test users' performance in six different factors: static-dynamic head movement, static-regular dynamic sound, and static-irregular dynamic sound. Based on the test results, we reflect and propose 3 strategies for user behavior design and spatial sound dynamic design. Additionally, we propose a sound effect design strategy as an additional contribution based on player behavior observation and interview results.

2 BACKGROUND AND RELATED WORK

2.1 VR Audio Game and Distance Perception

VR games that focus on spatial audio design have many different themes and styles. BlindSide is an audio adventure game with a horror theme that simulates the experience of blindness¹. Players navigate and solve puzzles in the virtual environment by perceiving sound distance through spatial audio. Daydream is a healing-style sound interactive adventure game that helps players slow down and relax through pleasant environmental sounds and sound effects [18]. In addition to enhancing immersion through spatial audio, Dark Echo is a game that uses audio emission and reflection positions for interaction². Eyes-Free Art is not a game, but an interactive installation designed for visually impaired people to interpret 2D

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¹2012 BlindSide <http://www.blindsidegame.com/>

²2015 Dark Echo https://store.steampowered.com/app/368650/Dark_Echo/

art using proxemic audio. Blind people can hear different sound explanations at different distances from the artwork [1].

2.2 Distance compensation in VR

There are many compensatory methods for distance perception in VR, mainly relying on visual, multi-channel sensory, audio, and reverberation [17]. Most methods rely on visual intervention, such as reducing visual images [20], reducing the scale of virtual space [23], and compensating for distance compression by adjusting the user's eye height for spatial visual calibration [22]. There are also works that use audio to compensate for distance compression, mainly by enhancing perceptual ability through the combination of multi-sensory information, such as verbal correction outside the field [26], and inconsistent audio-visual interactive combinations [8, 32]. Researchers have compared the compensatory effects of visual and auditory under the same conditions, and found that audio has better compensatory effects [14], because audio feedback usually has shorter interaction time than visual feedback [33]. Reverberation is an important knowledge in spatial acoustics [5, 12, 19], and is also an important strategy for distance compensation [2]. Reverberation has been shown to directly affect the perceived size of space [12], and there are works that study better distance compensation by controlling the parameters of reverberation [13, 14]. Audio design and experimental design are complex and rigorous. There are still many works on distance compensation, and the research community has not yet reached a clear consensus on the compensatory effects of different sensory and external stimuli in different spatial distance ranges. We attempt to contribute to this classic problem in the knowledge base through a small-scale design.

3 GAME CONCEPT DESIGN

"Sleep No Mosquitoes" is a VR game that runs on the HTC Vive headset. The game was developed using the Unreal Engine 4 game engine. Considering that the game is primarily sound-based, the Google Resonance Audio SDK was used to simulate the sound field as accurately as possible, effectively reproducing the reflection characteristics of sound in real space.

The HRTF (Head-Related Transfer Function) stands as a pivotal method in the realm of spatial sound simulation, particularly within the confines of VR gaming platforms, as it significantly amplifies the fidelity of the gaming content design. HRTF epitomizes an advanced technique for the real-time rendering of spatial sound by harnessing the power of algorithmic simulation in concert with head tracking technologies [6]. It encapsulates the acoustic impact of our head, including the modulation of sound signal phase and frequency, as sound waves traverse towards our ears.

Functionally, our ears act as dynamic, direction-sensitive sound receptors, registering auditory signals in an ongoing state of mechanical flux. HRTF technology is crafted to employ software or algorithms capable of pinpointing the spatial coordinates of these "sensors" with remarkable precision. By leveraging this technology, it is feasible to accomplish the real-time rendering of sound originating from various vectors, thereby emulating the sonic landscape of the real world and facilitating the phenomenon of audio spatialization. This level of acoustic verisimilitude is indispensable for crafting engrossingly authentic virtual experiences.

3.1 Game Overview

Embarking on a foray into "Sleep No Mosquitoes," you find yourself in the shoes of the protagonist during the sultry summer twilight of an urban setting. As the night's calm stretches before you, your slumber is rudely fractured by the incessant buzzing of mosquitoes. These vexatious intruders encroach from every conceivable angle, their droning heralding not just annoyance but the irksome pinpricks of their bites.

In this nocturnal skirmish, you're rendered powerless but not defeated—roused from your bed, you engage in a visceral conflict. With nothing but your hands as your weapons, you grapple in the twilight, silhouetted by the scant glow of the city night, as you strive to vanquish your tiny adversaries. It's a battle waged in the shadows, with each clap of your hands both a chance for victory and a reminder of your precarious situation in the onslaught of these relentless pests.

3.2 Game Play

We invited 21 participants to experience the game. During gameplay, participants must don VR gear and be seated in a designated rotating chair to fully immerse themselves in the virtual environment. They must use acute auditory perception to detect targets. Upon locating a mosquito, players extend their arms, press the controller button to mimic grabbing, and simulate squashing the mosquito as they would in reality.

The game is structured into three distinct levels, segmented into player engagement, enemy dynamics, and player-enemy interaction. Players, equipped with a VR headset, must hone in on auditory cues signaling the presence of virtual insects. Accurate auditory localization and prediction of flight patterns are crucial for timely capture of the enemy. Successful interceptions are confirmed by both visual and auditory feedback from the system.

During player-enemy encounters, haptic feedback from the VR controller signals the presence of a mosquito on the avatar. Players must act swiftly, shaking their hands vigorously to fend off the insects. Failure to do so will result in a decrement of the player's lifeline and, ultimately, the end of the game (as depicted in Fig.1). The description may benefit from enhanced clarity by integrating specific details regarding the visual and auditory cues and by explicitly stating how these cues enhance the gaming experience.

3.3 Game Level Design

The game's auditory landscape is meticulously crafted, with particular attention to mosquito sound design, tailored to each of the three distinct levels. These levels vary in mosquito behavior and attack patterns.

Each level opens with a practice session that acclimates players to the mosquito's buzzing, using a prominent particle visual effect for guidance—this aid vanishes in the subsequent challenge phase. Here, players find themselves in a dim setting, reliant solely on the mosquito's hum to track and capture them. Successive mosquito encounters occur within the level, culminating in a performance summary that highlights the time taken for each capture.

Level Descriptions:

- Level 1: Features a consistent mosquito buzz, enabling players to use directional sound to pinpoint and capture the insect.

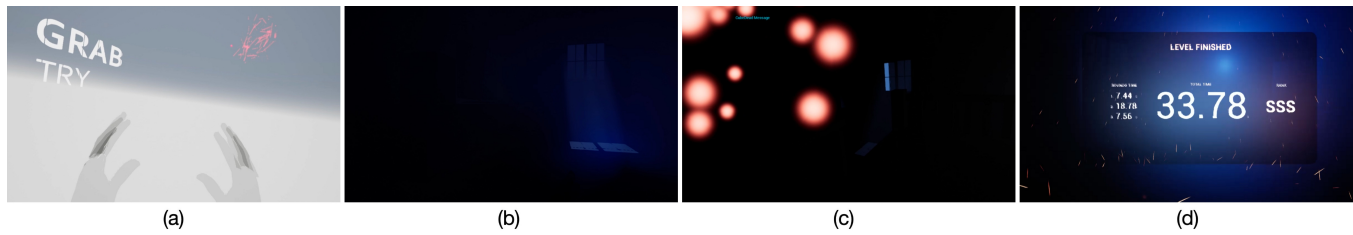


Figure 1: (a)Tutorial level. (b)Scene entering for the first time. (c)Feedback from successful mosquito bites. (d)Settlement interface.

- Level 2: Introduces mosquitoes on a predictable path around the player, allowing for sound-based distance estimation without head movement.

- Level 3: Presents erratic mosquito movement patterns, punctuated by brief pauses, challenging players to adapt and seize the opportunity to react.

The game design intends to escalate the player's competitive drive through these levels. Each mosquito represents a phase of the gaming experience: the first for hypothesis and exploration, the second for method validation and adjustment, and the third for applying knowledge to demonstrate mastery.

Additionally, an in-game aid offers audio cues correlating with arm movement. Using a diatonic scale starting at middle C, the pitch ascends as players' arms open, with the tempo of the accompanying sound matching the speed of the arm extension.

For game refinement purposes, a testing mode was developed, distinguishing itself from regular gameplay by allowing for manual audio control, head movement adjustments, and customizable test iterations.

3.4 Why mosquito?

The mosquito's buzz is irritating for three primary reasons: its frequency, the associated acoustic phenomena, and the impact of psychological factors [16]. The buzzing, predominantly a result of rapid wing flapping, typically falls within the 300-400Hz audio range. Studies on the audibility threshold curve indicate that humans can detect even low sound levels in this range, to which they are highly sensitive.

The mosquito's swift flight near the ear produces a pronounced acoustic sensation known as the Doppler effect. As mosquitoes approach, the sound waves compress, increasing frequency and creating a sharper sound; conversely, when they move away, the waves elongate, decreasing frequency and resulting in a duller sound. This fluctuation in pitch not only heightens annoyance but also reveals the insect's proximity (the Doppler effect additionally allows for the perception of object distance).

Psychoacoustics, the study of sound perception, suggests that psychological aspects are crucial to our response to mosquitoes' buzzing. Individuals with a history of bites may associate the sound with the anticipation of the discomfort and itching that follow, magnifying their aversion to it. This psychological response not only intensifies the annoyance but also triggers an instinct to swat at the sound source. Consequently, the irksome nature of mosquito

buzz is leveraged in games to provoke player engagement, eliciting instinctive physical responses akin to swatting the pests.

3.5 Game Features

VR Audio Field Reconstruction

To faithfully replicate auditory experiences, it is imperative to account for sound's physical attributes. Specifically, the depiction of a mosquito's flight demands the incorporation of the Doppler effect to augment the sound's spatial and dynamic qualities. Unreal Engine 4 and Resonance Audio are employed to construct the game's acoustic environment. Within this setting, it is critical to define the sound's trajectory, attenuation curve, and precise Doppler effect parameters. Additionally, to enhance the realism of the mosquito's buzz, the source volume and pitch should fluctuate randomly within a predefined spectrum.

Psychoacoustics in Game

In game audio design, psychological factors also play a significant role. Psychoacoustics tells us how the human brain processes sound. The way people react to sound is closely related to their subjective factors.

For example, an individual frequently bitten by mosquitoes; they might link the buzzing to sensations of fear and itchiness. This association can intensify their dislike for the sound and provoke an instinctive swatting response. Utilizing mosquitoes as the primary antagonistic sound in games capitalizes on this aversion, promoting physical engagement from players. In generating the sound source, the search for authentic mosquito audio often leads to film and TV databases. The creation process involves selecting various natural mosquito sounds, which are then modified using audio editing software to emphasize high-frequency elements. After multiple rounds of A/B testing, the most suitable mosquito audio for game integration is selected.

Atmosphere Design

Researchers acknowledge that in the use of audio compensation or reverberation compensation, as well as in multi-sensory combination compensation, visual capture exists, although vision dominates in creating perception [3, 11]. However, visual blur has no significant effect on distance perception [21]. Therefore, in game scenes, players can only see faint light inside the house. Players can locate themselves in space through blurry visual clues. In addition, the pitch-black environment provides players with a certain visual barrier without making them feel too confined in the immersive VR environment. By designing dark and blurry scenes, players are encouraged to enhance their imagination of the game world.

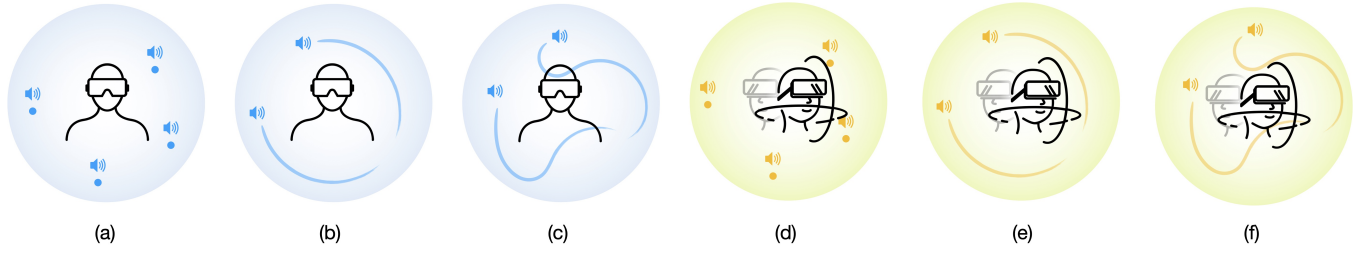


Figure 2: Figure 2 illustrates the six experimental stimuli used in the study: (a) features a stationary head position with a constant sound; (b) maintains a stationary head position paired with a consistent, changing sound; (c) also has a stationary head position, but is coupled with an inconsistent, varying sound; (d) contrasts with a moving head position while the sound remains constant; (e) combines a moving head position with a consistent, changing sound; (f) involves a moving head position and an inconsistent, varying sound. This figure succinctly categorizes the stimuli into various combinations of head movement and auditory dynamics to assess the test subjects' responses under different conditions.

In games, some action sound effects are particularly important, including: hitting sound effects, missing sound effects, the rustle of clothing from limb movements, victory cheers sound effects, and text and number prompt sound effects. The hitting and missing sound effects are related to the player pulling the trigger button, which is related to the action held by the player's hand. We designed the action sound effects of the auxiliary character reaching out to help players perceive the distance between their outstretched arm and the target object.

4 GAME TESTING

This experiment utilizes a quantitative approach, carried out in a controlled game testing environment comprising multiple trials under defined conditions.

4.1 Parameters and Procedure

The purpose of the game test is to evaluate whether the motion of the head and the dynamic design of the audio have a positive compensatory effect on VR distance compression. We conducted 6 experiments comparing two types of head motion (see Fig 2): static and free dynamic. Three types of sound motion were also compared: static, regular dynamic, and irregular motion. The distance compensation effect was quantified by measuring the time it took for participants to hit mosquitoes. The experiment was conducted in a controlled laboratory environment using HTC VIVE as the HMD. Mosquito movement was controlled within the arm span of each participant. Participants wore noise-cancelling headphones to isolate environmental noise in the real space.

A total of 10 participants were recruited (male = 5, female = 5, mean age = 28.6, $SD = 10.1$). After a brief informed consent, we invited participants to wear equipment, set up the experimental environment, and provide a simple game to guide participants to understand the experimental requirements and VR environment. Each participant completed 6 experiments in a 2×3 subject design, and each experiment required the participant to hit 10 mosquitoes and record the time taken for each hit. After completing one experiment, the next experiment was conducted directly, for a total of 60 hits. The order of the 6 experiments was balanced using a Latin square. User behavior was recorded by cameras outside the field,

and a principal investigator was also present to record participant behavior at any time.

The experiment assessed participants' proficiency in gauging distances with the use of time as a measurable indicator. This decision was driven by two primary considerations: firstly, the time taken to complete tasks could reliably indicate the rate of successful outcomes, which in turn, would facilitate the evaluation of whether diverse experimental approaches improved the accuracy of distance perception. Secondly, employing time as a key determinant of game results was intended to invoke a competitive spirit and challenge-seeking behavior among the participants.

4.2 Results

In order to determine whether different types of dynamic head and audio displays affect participants' perception of spatial distance, we encoded participants' responses as binary variables based on task time and conducted an analysis of variance (ANOVA). The results showed that whether it was head movement ($F(1,594) = 22.90, p < 0.001$), audio changes ($F(2,594) = 255.30, p < 0.001$), or both head and audio factors simultaneously ($F(2,594) = 61.79, p < 0.001$), all significantly affected task time. Specifically,

1. Compared to static head position (17.10+-13.64 seconds), completing the task with dynamic head movement resulted in shorter task completion times (14.29+-5.17 seconds), indicating that head movement can compensate for users' perception of spatial distance.

2. Compared to static audio (17.59+-12.05 seconds), completing the task with dynamic audio resulted in shorter task completion times (12.61+-5.38 seconds), indicating that dynamic audio changes can also compensate for users' perception of spatial distance. Since a significant interaction effect existed between Head and Sound, we then performed simple effects tests and found that Time was significantly different between the two Head levels at irregular sound dynamics ($F(1,198)=8.19, p<0.001$), and also significantly different at regular sound dynamics ($F(1,198)=11.085, p<0.001$) and static sound ($F(1,198)=60.05, p<0.001$). Importantly, we found that:

3. In the case of regular audio dynamics, completing the task with static head position resulted (9.64+-4.10 seconds) in shorter task completion times than with dynamic head movement (11.48+-3.70 seconds) ($F(1,198)=11.09, p=0.001$), indicating that simultaneous



Figure 3: (a-c) Participants' test scenarios. (d) Sound effects of body movement.

dynamic head and audio displays may not effectively compensate for spatial distance perception.

4. In the case of irregular audio dynamics, completing the task with static head position resulted (10.56±5.16 seconds) in shorter task completion times than with dynamic head movement (12.33±3.45 seconds) ($F(1,198)=8.19$, $p=0.005$), indicating that simultaneous dynamic head and audio displays may not effectively compensate for spatial distance perception.

Post hoc pairwise comparisons were further conducted. The results indicate that:

5. Under dynamic head movement conditions, there was no significant difference in task completion time between regular audio dynamics ($p=0.275$). Similarly, under static head position conditions, there was no significant difference in task completion time between irregular audio dynamics ($p=0.771$).

4.3 Investigation and Findings

This section presents observational data on the behavior 10 participants in experimental and 21 participants in standard gameplay contexts (see Fig.3(a-c)).

Initially, some participants required time to acclimate to the VR technology. Once accustomed to the gameplay, they entered a state of flow, allowing them to fully engage with the sound design and enjoy the VR gaming experience.

The mosquito-catching game drew significant interest from participants, who eagerly interacted with the virtual mosquitoes and remained highly focused. Over half the participants recalled memories of sleepless nights disturbed by mosquitoes, contributing to a profound sense of immersion. Swatting mosquitoes in VR also provided a sense of catharsis.

In the standard game mode, there was a pronounced desire to succeed among players. They expressed eagerness to replay the game and improve their completion times. This competitive drive was particularly evident in level three, where irregular mosquito sounds increased user interaction. Players felt a strong sense of accomplishment after finishing the game and were keen to discuss their strategies with others. Notably, level three experienced the highest replay rates.

As for in-game assistive features, the arm-stretching sound effect garnered unexpectedly favorable reactions. While formal user research was not conducted, feedback from participants indicated that the audio cue helped them judge the distance to targets and the speed of their arm movements, proving especially helpful in low-visibility VR scenarios (see Fig.3(d)).

In the game's preliminary stages, we developed audio for various insects—including mosquitoes, flies, and bees—with specific

frequency adjustments to mimic different species and habitats. During pre-testing of these audio samples, user interviews revealed that the sound of mosquitoes instinctively prompted them to react, with higher-pitched tones drawing more attention.

5 DISCUSSION AND DESIGN STRATEGIES

In VR, compensating for distance compression through multi-channel compensation is an area of expertise in the audio engineering field. This work combines the design experience of previous work in compensation strategies for vision, hearing, and other areas [17, 20, 26], and attempts to propose three new design strategies through a fun mosquito-catching game design and testing to compensate for the problem of distance perception compression in VR lightweight games. These strategies may only be applicable to related game styles and game volumes.

From the experimental data, it can be seen that under two different audio dynamic conditions, compared to head dynamics, the task completion time is shorter when the head is static. The reasons for this may include: (1) when there is sound or head dynamics, it satisfies the state of relative motion of the sound, which is more beneficial for participants to locate sound. (2) According to the observation, participants rarely reach out while their heads are in motion, which may not be in line with natural human behavior. For the new findings revealed by the experimental results, we look forward to further verification in future work.

Enhancing Dynamic Audio in VR.

Our VR game design investigation uncovered that dynamic audio plays a pivotal role in improving users' spatial distance perception over static audio. Dynamic audio heightens user attention, sharpens alertness, and influences behavior. In the virtual realm, dynamic audio not only informs about the sound source but also accentuates other sound properties like location, scale, and the intricacy of the transmitted information.

Incorporating Dynamic Head Movements

Traditional non-VR games restrict players to controlling a character's orientation to perceive audio sources, leading to a condensed perception of sound information. However, applying knowledge of the Head-Related Transfer Function (HRTF), which governs spatial audio localization and distance perception, VR gameplay can be enhanced. By integrating head movements with the positioning of external audio sources, we encourage players to refine their auditory direction and distance awareness, aligning closer to real-life interactions and increasing immersion.

Integrating Avatar-Generated Sound Effects

Simulating limb movement sounds, akin to those made by real-world arm stretching, significantly enriches the player's sensation

of arm extension and spatial distance in VR. While noise-cancelling headphones are common in VR to foster immersion by cutting off real-world sounds, they exclude environmental noises like limb friction. By reintroducing these sounds as in-game elements, we offer a potential solution to the issue of distance perception compression in virtual environments.

6 CONCLUSION

This Sleep No Mosquitoes is a game design exploration. By metaphorically placing the real-life scenario of swatting mosquitoes into a virtual environment, we have discovered some designs that can compensate for the distance compression problem in VR. We have designed both entertainment and experimental modes for the game, and from quantitative testing, we have found that dynamic head movements and audio can significantly improve users' perception of distance in VR. Additionally, from observing players' performance and conducting interviews, we have found that adding sound effects related to body movements can also be effective in compensating for the problem. We hope that this design will provide inspiration for addressing the spatial distance compression issue in VR environments.

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